The empirical recurrence for OEIS A253699, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

5 September 2026

Abstract

OEIS A253699 counts arrays of width 3 over $\{0, 1\}$ in which two diagonal numbers of every 2×2 subblock each differ from the same number at a neighbouring subblock. The entry carries an empirical recurrence of order 16 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: a block of k rows and the block one row lower together occupy $k + 1$ consecutive rows, so every comparison the condition makes is settled inside 3 consecutive rows and the count is a walk count on $S = 49$ states.

2020 Mathematics Subject Classification. 05A15, 05B20, 15A18.

1 The conjecture

OEIS A253699 is "Number of (2+1)X(n+1) 0..1 arrays with every 2X2 subblock ne-sw antidiagonal difference unequal to its neighbors horizontally and nw+se diagonal sum unequal to its neighbors vertically.". It has offset 1 and begins

$$40, 76, 150, 282, 540, 1026, 1984, 3782, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 4*a(n-2) -2*a(n-4) +5*a(n-5) -7*a(n-6) -3*a(n-7) +9*a(n-8) -5*a(n-9) +a(n-10) +5*a(n-11) -6*a(n-12) -2*a(n-13) +4*a(n-14) -a(n-16)

As of the "Last modified" line on the live entry (Jul 23 2025 14:10:15, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 3 columns and a growing number of rows, with entries in $\{0, 1\}$.

The entry states two conditions at once. For a $k \times k$ block B of cells of the array put

$$f(B) = B_{1,k} - B_{k,1} \qquad \text{compared vertically,}$$

that is, the northeast entry minus the southwest entry;

$$g(B) = B_{1,1} + B_{k,k} \qquad \text{compared horizontally,}$$

that is, the northwest entry plus the southeast entry. Each condition says that its own number takes different values on two blocks that are neighbours in its own directions: with $B(i, j)$

the block whose top-left cell is (i,j) and (δ_1,δ_2) the step for the direction concerned — $(1,0)$ vertically, $(0,1)$ horizontally —

$$f\big(B(i,j)\big) \neq f\big(B(i+\delta_1,\, j+\delta_2)\big) \quad \text{resp.} \quad g\big(B(i,j)\big) \neq g\big(B(i+\delta_1,\, j+\delta_2)\big),$$

whenever both blocks lie wholly inside the array. Both conditions are imposed for that one size of block, 2×2. The entry writes the array with n as the number of COLUMNS. Transposing it exchanges the horizontal and vertical directions and fixes the two diagonal ones, and it either fixes the number f or replaces it by its negative — the northeast and southwest entries change places. An equality is unaffected by negating both sides, so the transposed problem is the same problem with the horizontal and vertical directions exchanged, and the directions above are named in the transposed frame.

Two remarks fix the reading. A neighbouring block is the block one step over, overlapping the first, not the next disjoint block: the entry says "its neighbors", and the counts below confirm it. And a comparison is imposed only when both blocks lie wholly inside the array, so a block near an edge is compared in fewer directions and a block whose named neighbour falls off the array is under no condition in that direction.

3 A window of 3 rows

A $k\times k$ block occupies k consecutive rows, and each of its neighbours occupies either the same k rows or the k rows one lower. So both blocks of any comparison lie inside $k+1$ consecutive rows, hence inside 3 consecutive rows for every size the entry names. No comparison reaches further than that, in either direction.

Take the array one column at a time and let the state be the last 2 rows written so far, with a distinguished start standing for the array before its first row, so that a walk of n steps is an array of n rows. Appending a row produces a window of 3 rows, and the step is allowed exactly when every comparison whose two blocks both lie inside that window holds.

Every comparison of the finished array is tested this way. Each one lies inside 3 consecutive rows, and every 3 consecutive rows of the array form the window of some step. A comparison lying inside more than one window is tested more than once, which costs nothing. Conversely no step imposes anything not asked by the definition, since a step tests only comparisons whose two blocks lie inside the array.

Every state may end an array, so $\tau = \mathbf{1}$. A block sitting in the final rows whose neighbour would lie below the last row is not compared in that direction at all — the neighbour is not part of the array — so stopping never leaves a condition unchecked. For the same reason the array of exactly 2 rows is handled by the walks of length 2 with no further test: all of its comparisons already lie inside the windows of those steps.

Thus

$$a(n) \;=\; \iota^\top M^j \tau, \qquad S = 49,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^{16} - \sum_i c_i t^{16-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+16} - \sum_i c_i M^{j+16-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) \;=\; 4\,a(n{-}2){-}2\,a(n{-}4){+}5\,a(n{-}5){-}7\,a(n{-}6){-}3\,a(n{-}7){+}9\,a(n{-}8){-}5\,a(n{-}9){+}a(n{-}10){+}5\,a(n{-}11){-}6\,a(n$$

for every $n > 16$.

Proof. The vector $w = q(M)\tau$ was formed by 16 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 16 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 29 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of width 3 over $\{0, 1\}$ with a few rows was written out cell by cell, every block of that one size was located, its number computed, and the comparison made with each neighbouring block that lies inside the array. No window, no state and no end vector are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A253699.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.